Crop Type Classifiication – Project Report

We built a pipeline that identifies the crop in a farm from Sentinel-1 and Sentinel-2 time series; the first version classifies 61–63% of unseen farms correctly across 10 classes.

Objective and data

The goal is to identify the crop grown in each farm polygon from satellite imagery, using a per-crop "signature" of spectral bands, vegetation indices and radar backscatter.

- **Ground truth:** 5,910 surveyed farm polygons with GPS points (shapefile of 27 Oct).
- **Season used:** Rabi 2022-23, which holds 89% of the labels. Other seasons have too few fields to train on.
- **After quality filters:** 5,188 fields in 10 classes: Wheat 1,246, Gram 1,047, Onion 774, Sorghum 544, Maize 467, Other 390, Cotton 324, Sugarcane 273, Soybean 77, Brinjal 46.
- **Split:** the shapefile's own field-level split, with 3,584 fields for training and 1,604 for testing.

Method

Each crop is told apart by how it changes over the season, not by how it looks on one date.

1. **Sampling:** for each field, two ways of choosing pixels away from mixed edges were compared.
    a. The largest square inside the field after a 10 m inner buffer (median 4.6 pixels).
    b. A 30 × 30 m square around the surveyor's GPS point, clipped to the field (median 8.4 pixels).
2. **Satellite features:** Google Earth Engine composites every 15 days from 1 Oct 2022 to 6 Apr 2023 (13 dates):
    a. Sentinel-2: 12 bands and 6 indices (NDVI, EVI, NDMI, NDRE, SAVI, GNDVI), with clouds removed using Cloud Score+.
    b. Sentinel-1: VV, VH, the VH/VV ratio and the radar vegetation index (RVI).
3. **Gap filling:** dates hidden by cloud or missed by the radar are filled by linear interpolation in time.

4. **Signature:** each field becomes 298 values (22 features × 13 dates, plus 12 phenology metrics such as peak date and season length). Each crop's signature is the median over its training fields.
5. **Models:** nearest-signature matching, Random Forest and XGBoost, with class weights so small crops count as much as large ones.

Everything runs with one command: `python run.py`.

Results

XGBoost is the best model; the GPS-point square edges out the polygon square on accuracy and on unseen villages.

Sampling	Model	Accuracy	Macro F1	Macro F1, unseen villages
GPS-point square	XGBoost	0.63	0.50	0.51
Polygon square	XGBoost	0.61	0.50	0.48
Polygon square	Random Forest	0.59	0.47	–
GPS-point square	Random Forest	0.59	0.46	–
Either	Signature matching	0.42	0.33	–

Scores are on the 1,604 test fields. Macro F1 averages all 10 crops equally. The unseen-village score holds out whole villages, the best guide to performance in a new area.

- **Recognised well (recall 64–75%):** sugarcane, gram, onion, cotton, wheat.
- **Often confused:** maize and sorghum, mostly predicted as gram or wheat.
- **Rarely recognised:** soybean and brinjal (11% and 7% recall).

Key fiindings and limitations

- **Seasonal timing separates crops.** Onion peaks late (Feb–Mar), gram dries down first, sugarcane and cotton are already green in October.

- **Wheat, gram, sorghum, maize and soybean have similar curves**, all peaking in Dec–Jan. This sets the current accuracy ceiling.
- **Red-edge and soil-adjusted indices matter most** (NDRE, SAVI, EVI, NDVI), followed by the shortwave infrared bands B11 and B12.
- **More pixels per field helps.** Many fields are narrow strips, so the inner square holds only about 5 pixels. The larger GPS-point square scores higher.
- **Signatures explain crops but classify poorly.** Matching a field to the nearest crop signature reaches 42%; machine-learning models using the same signatures reach 63%.
- **Data limits:** only one Sentinel-1 satellite was flying in 2022-23, so 34% of field-dates lack radar. Soybean is a kharif crop, so its 77 rabi labels are probably wrong.

Next steps

Planned improvements to accuracy, in order of expected gain:

- [ ] Sample the whole field minus a 10 m edge (about 17 pixels instead of 5).
- [ ] Train on individual pixels, then combine them into one prediction per field.
- [ ] Drop or merge the soybean and brinjal classes, and review doubtful labels.
- [ ] Add more dates: 10-day composites, or a window starting in September.
- [ ] Build a separate model for Kharif 2026 (376 fields) once more labels arrive.